Table 4-1. *Comparative Operators*

Symbol	Description	Is true if…
==	is equal to	the left side is equal to the right side
!=	is not equal to	the left side is not equal to the right side
<	less than	the left side is less than the right side
>	greater than	the left side is greater than the right side
<=	less than or equal to	the left side is less than or equal to the right side
>=	greater than or equal to	the left side is greater than or equal to the right side

Comparative operators work great if needing to compare two values against each other, but you might need to make some more complex logical comparisons. In this way, you could use a **logical operator**, for example, to make two separate comparisons, and only do something if both comparisons are true. Consider the following example expression:

```
myValue >= 10 && myValue <= 20
```

This expression uses the **logical and** (&&) operator, returning true if and only if the conditional statements on both the left and the right are true. In this case, this statement will be true if `myValue` contains a value greater than or equal to 10 and less than or equal to 20. This creates a range of numbers between 10 and 20 that could be true.

Conversely, the **logical or** (||) operator will return a value of true if either of the conditional statements on the left or right is true. Take this following example:

```
myValue < 10 || myValue > 20
```

This line would only return true if `myValue` contained a value that was less than 10 or greater than 20, excluding the values from 10 through 20.

The final logical operator is the **not** (!) operator. This one is a little different from the others in that it works on only one operand and returns true only if the expression is false. Take the following example:

```
!myValue
```

This line will return true only if `myValue` is false or contains the numerical value of 0.

Comparative and logical operators are going to play a crucial part in the Arduino decision-making process. Table 4-2 provides a quick reference of the logical operators to refer back to.